

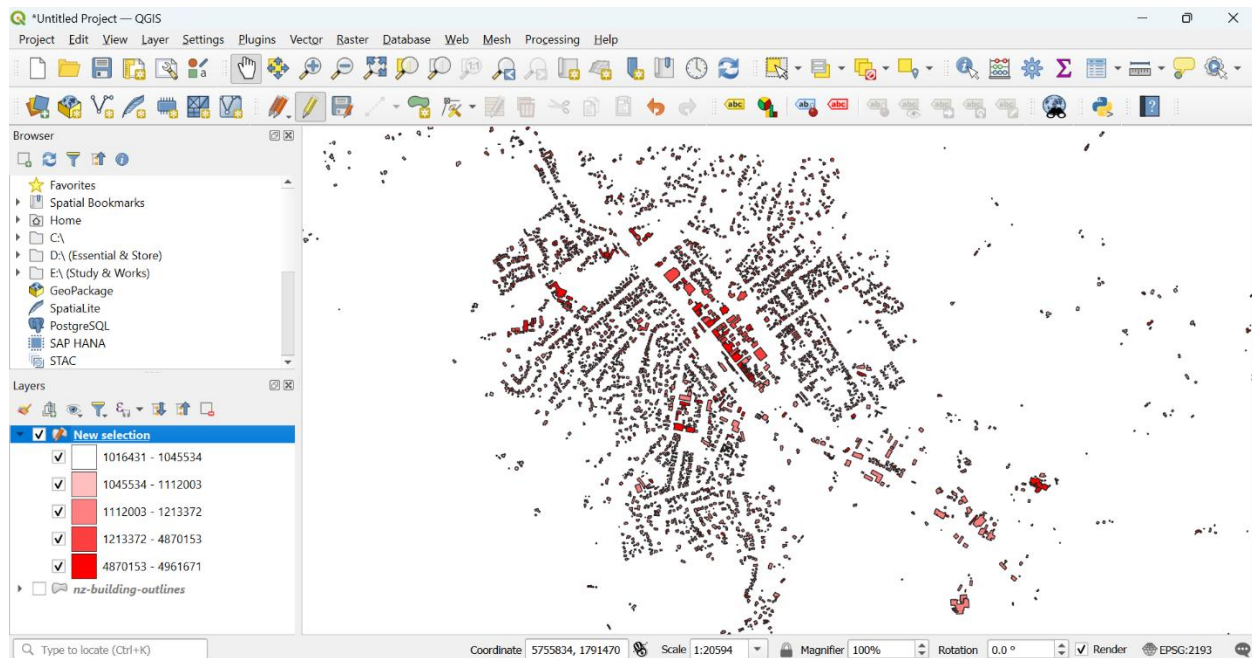
At-Home Activity 1: Choropleth Mapping

Dataset: **NZ Building Outlines** <https://data.linz.govt.nz/layer/101290-nz-building-outlines/>

I selected a portion of the map to calculate easily in short time using the “Select Features By” option and saved as name “New selection” in QGIS.

a. Which attribute did you choose to map and why?

I chose the "building_i" that is for Building ID attribute from the NZ Building Outlines dataset for this choropleth mapping. This attribute represents unique building identification numbers, which can be useful in analyzing the building area and density in different areas. By mapping this attribute, we can visualize patterns of building concentration across a region.



b. How did you calculate the values for the normalised versions of the map?

I calculated two new attributes that is Area and Density by using Field Calculator in QGIS as they were not present in the dataset and then created separate choropleth maps for each of the attributes as below:

Deriving the Area

I calculated the area for each building polygon using the Field Calculator in QGIS:

- Attribute Table > Field Calculator > new field "Area".
- Used the area(\$geometry) function to compute the area of each building.

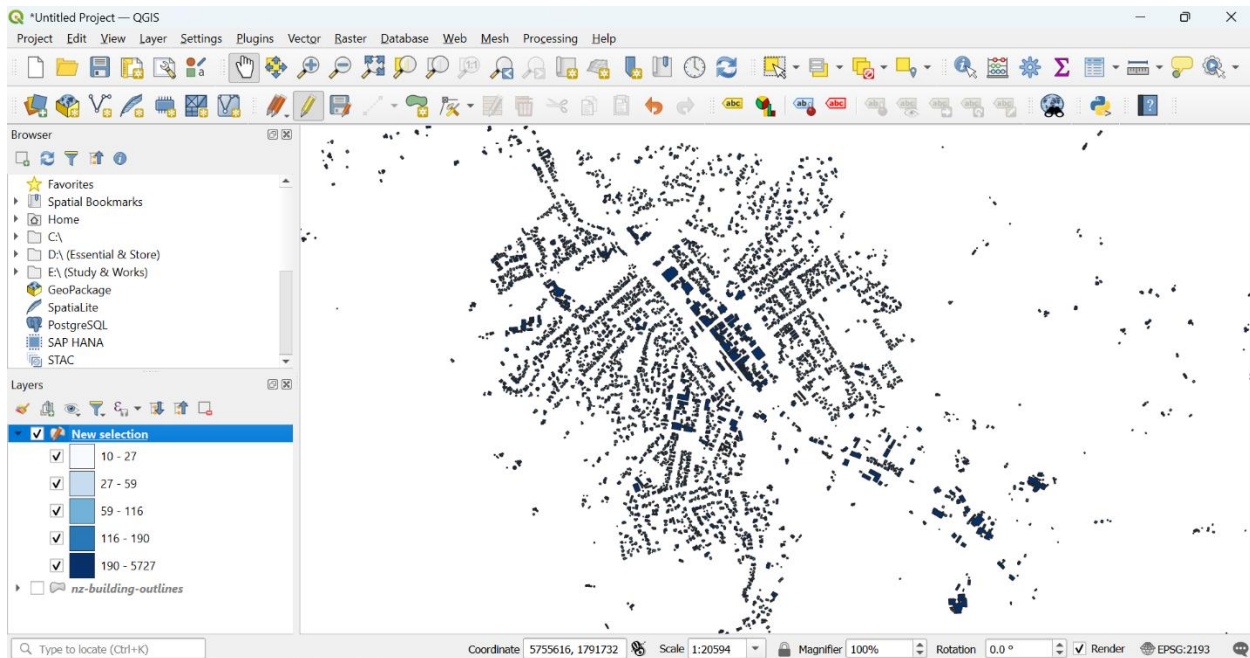
Deriving the Density

Then, I calculated the building density, which represents the number of buildings per unit area:

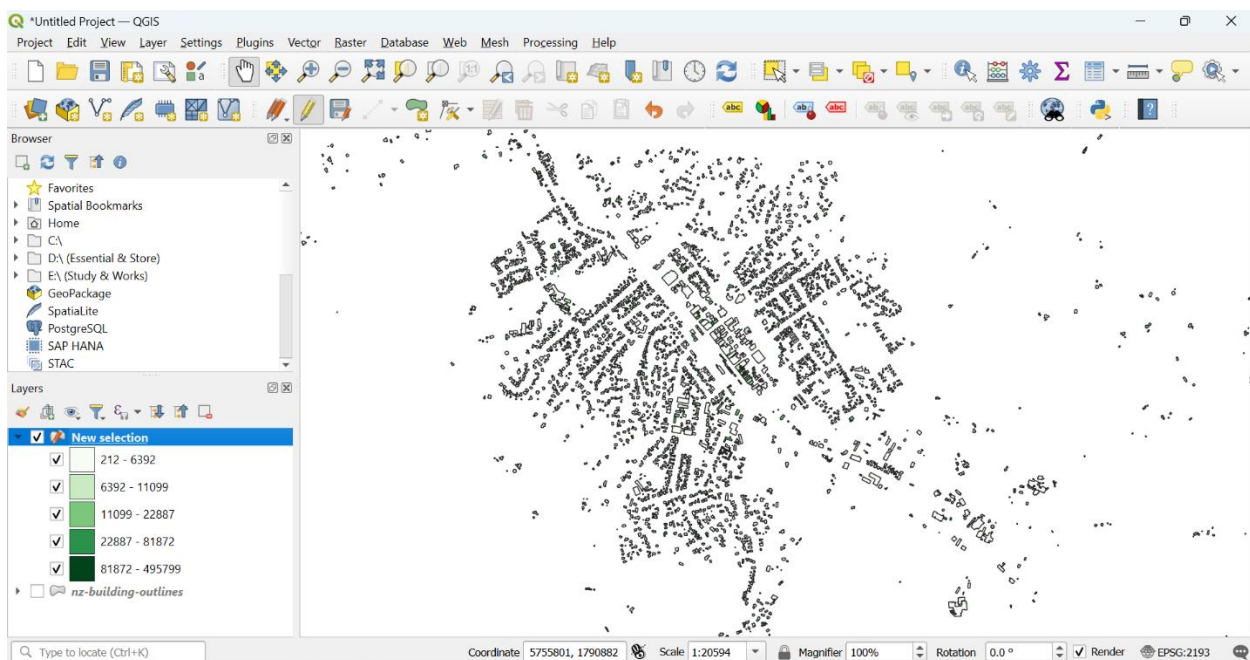
- Attribute Table > Field Calculator > new field "Density".
- Used the formula: "building_i" / "Building_Area" to get the density of buildings.
- A scaling factor (* 10,000) was applied to make the values easier to interpret.

Creating Choropleth Maps

After deriving these attributes, I created separate choropleth maps: One map based on Area to visualize the size distribution of buildings. I used blue color scheme this time to visualize the area.



And another map based on Density to show variations in the concentration of buildings across different regions. This time I used green color to visualize.



These normalized maps provided better views by adjusting for area differences, allowing for a more meaningful comparison between regions.

c. Why did you select those normalised versions, and what differences can you see between the three maps?

I selected Building Area and Building Density as normalised versions because they can provide more meaningful information about the building in any area. I can compare now building distributions in any region by these normalised versions. Just the count of buildings does not provide that meaningful information of any region, they only shows the number of building distributions. Normalising the data now ensures that the choropleth maps are reflecting more meaningful information rather than just showing the count.

The main difference between these three maps is that, the first map is just showing the building count that does not show the variations but the second map is reflecting the physical size of each building that is now more useful visualization. While the third map is demonstrating the number of buildings per unit area. So it can be understood that how densely a region is built-up by the map.

d. Which of the three maps is most informative, and why?

Among the three maps, the third map that is Density based map, that is most informative because it holds the most useful information how many buildings are there in a particular region. It reflects how a region is built-up and how is the density of building and population. That can be used for urban planning and development of a region.